

Universite Paris Dauphine, PSL

Master 1 MIAGE

Solving the Rubik's Cube

with the A* Search Algorithm

Course: Artificial Intelligence and Reasoning

Lab 1, Project Report

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Abstract

This report presents a program that solves the Rubik's Cube. We model the cube as a search problem and we solve it with the A* algorithm seen in Lecture 2. We first run a uniform cost search. We then add two admissible heuristics that guide the search. The first heuristic relaxes the problem so that each sticker moves on its own. The second heuristic relaxes it so that each piece moves on its own. We measure how scrambled a cube can be while still being solved fast. The results confirm the course: a good heuristic gives large savings, and A* runs out of memory long before it runs out of time.

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1 Introduction

A Rubik's Cube is a small cube. Each of its six faces holds nine colored stickers. Every face can turn, which mixes the colors. The cube is solved when each face shows a single color. Our goal is to write the "solve" button of a cube simulator. Given any mixed cube, the program must return a list of moves that brings it back to the solved state.

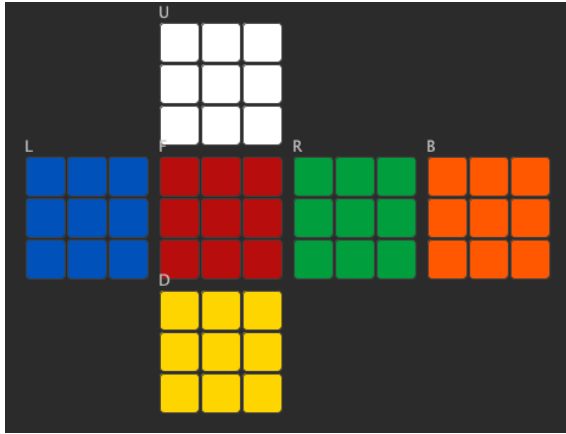
This problem fits the search framework of the course. A cube configuration is a state. A quarter turn of a face is an action. The solved cube is the goal. Every move costs one, so the shortest sequence of moves is the optimal solution. We therefore use the search methods of Lecture 2. In particular we use A*, which returns an optimal solution when its heuristic is admissible.

The report is organized as follows. Section 2 states the problem with the five components of a search problem. Section 3 describes the classes of the program. Section 4 explains A* and our implementation. Section 5 presents the two heuristics and shows that they are admissible. Section 6 reports the experiments. Section 7 concludes.

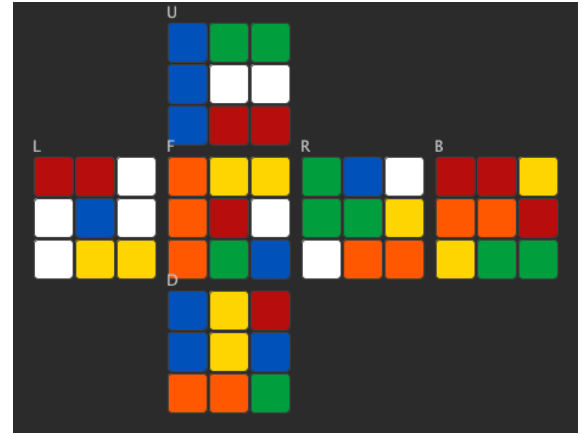
2 Problem formulation

2.1 The cube and its representation

We follow the Western color scheme. The solved cube shows white on the UP face, yellow on DOWN, red on FRONT, orange on BACK, blue on LEFT and green on RIGHT. Opposite faces are UP and DOWN, FRONT and BACK, LEFT and RIGHT. Each face is a three by three matrix of characters. The characters are 'W', 'R', 'G', 'B', 'Y' and 'O', as the statement requires. Figure 1 shows the solved cube and a scrambled cube drawn by our interface as an unfolded net.



(a) Solved cube.



(b) Cube after twelve random moves.

Figure 1: The cube shown by our interface as an unfolded net. The center of each face never moves, so it keeps its solved color.

Inside the program the 54 stickers are stored in one flat array. The matrix of a face is a view over this array. Each face uses row and column indices from 0 to 2, as in the figure of the statement. To turn a face we must know where every sticker goes. We place each sticker in a 3D frame, shown in Figure 2, where the cube is centered at the origin. A turn is then a real rotation of one layer around an axis. This way the permutation is computed, not typed by hand, so it cannot contain a wrong strip.

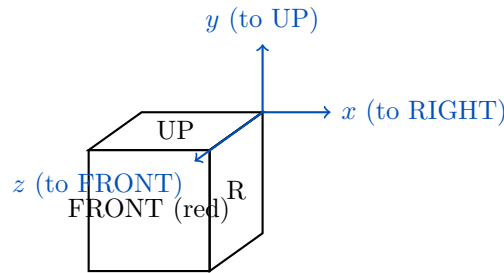


Figure 2: The 3D frame used to compute the turns. The outward normal of each face points along one axis. A clockwise turn rotates one layer by 90 degrees.

2.2 The five components

We define the search problem with the five components of Lecture 2.

- **Initial state.** The scrambled cube given by the user.
- **Actions.** The twelve quarter turns. Six are clockwise (U, L, F, R, D, B). Six are counter clockwise (U', L', F', R', D', B').
- **Transition model.** Applying a turn produces a new cube. The result of an action permutes the stickers of one face and its belt.
- **Goal test.** The cube is solved when every face is uniform with its solved color.
- **Path cost.** Each move costs one. The cost of a path is its number of moves.

The state space is huge. A real cube has about 4.3×10^{19} reachable configurations. We cannot store them all. A* must therefore explore as few states as possible, which is exactly why a good heuristic matters.

2.3 The twelve actions

Each action has an integer identifier, from 0 to 11, as in the statement. Table 1 lists them with their notation.

Id	Method	Notation	Sense	Id	Method	Notation	Sense
0	turnUp	U	clockwise	6	returnUp	U'	counter
1	turnLeft	L	clockwise	7	returnLeft	L'	counter
2	turnFront	F	clockwise	8	returnFront	F'	counter
3	turnRight	R	clockwise	9	returnRight	R'	counter
4	turnDown	D	clockwise	10	returnDown	D'	counter
5	turnBack	B	clockwise	11	returnBack	B'	counter

Table 1: The twelve actions and their identifiers.

2.4 Why this search is hard

The course measures a search by three quantities. The branching factor b is the number of children of a node. The depth d is the depth of the shallowest goal. The maximum path length is m . For our problem b is twelve, because each state has twelve children. The number of nodes generated by a blind search grows like b^d . With $b = 12$, even a small depth gives a very large tree. For example, 12^7 is more than thirty million. This is why a uniform cost search cannot go deep, and why we need a heuristic.

Measure	Symbol	Value for our problem
Branching factor	b	12 (twelve turns)
Worst case time	$O(b^d)$	exponential in the depth
Worst case space	$O(b^d)$	the real limit we hit

Table 2: Complexity terms of the course applied to the cube.

3 Program architecture

The code is written in Java. We chose Java because the statement describes a graphical interface, an interface for heuristics, and Java collections such as the linked list. The classes follow the statement closely. They are grouped by role, as Table 3 shows.

Package	Class	Role
model	Cube	Six face matrices, the twelve turns, isSolved
model	Face, Color	Faces, colors and their solved values
model	Move	Notation and inverse of each action
search	State	A node of the search tree, with expand
search	Astar	The A* algorithm and the solve method
search	StateSet	The explored set, with add and contains
search	PriorityQueueState	The frontier, with push and pop
heuristic	Heuristic	The heuristic interface, with value
heuristic	HeuristicLabel	The sticker relaxation heuristic
heuristic	HeuristicCube	The piece relaxation heuristic
ui	RubiksCubeSimulator	The interface and the solve button

Table 3: Main classes of the project, grouped by package.

3.1 The Cube class

The **Cube** class models the game. It stores the 54 stickers and offers the twelve turns and the **isSolved** test. The hard part is to know which sticker moves where for each turn. Writing these tables by hand is error prone. Instead, we give each sticker a position in 3D and a face normal. We then rotate the affected layer with a real 90 degree rotation. We read the permutation back from the rotated positions. This is correct by construction.

We checked the model with algebraic invariants. A turn followed by its inverse is the identity. The same turn four times is the identity. A turn and three turns the other way give the same result. The number of stickers of each color stays nine. The classic sequence R U R' U' returns to the solved cube after exactly six repetitions. All these tests pass.

3.2 The State class

A **State** is a node of the search tree. It stores the cube, the number of actions from the root (the cost g), the parent state, the action used to reach it, and the heuristic value h . The constructor computes h once. The **expand** method applies the twelve turns and returns the twelve children in a linked list, as the statement asks. Listing 1 shows the method.

Listing 1: The expand method generates the twelve children.

```
public LinkedList<State> expand() {
    LinkedList<State> children = new LinkedList<>();
    for (int action = 0; action < 12; action++) {
        Cube next = cube.withAction(action);
        children.add(new State(next, nbrActions + 1, this, action));
    }
    return children;
}
```

Each child keeps a reference to its parent and to the action that produced it. These two fields, called **pere** and **actionPere** in the statement, let us rebuild the full solution once the goal is reached. We compare two states by their cube only. Two paths that reach the same configuration are therefore the same state. This is what lets the explored set and the frontier remove duplicates.

3.3 The explored set and the frontier

The `StateSet` class is the explored set. It offers `add` and `contains`. It also keeps the smallest cost with which each configuration was expanded. This lets A* skip a configuration already reached by a cheaper path. The `PriorityQueueState` class is the frontier. It offers `push` and `pop`. The `pop` method returns the state with the smallest value $f = g + h$. The `push` method checks if the same configuration is already in the frontier and keeps only the better one.

3.4 The graphical interface

The `RubiksCubeSimulator` class is the interface. It holds a cube and draws it as the net of Figure 1. It has a `main` method that creates the window. Its `actionPerformed` method reacts to every button. There is one button per turn, a scramble button, a reset button, and the solve button. There is also a small menu to pick the strategy.

When the user presses solve, the program runs A* on a copy of the cube. The search runs on a background thread, so the window stays responsive. When the plan is ready, the interface plays it move by move on the cube, like a short animation. The log panel shows the plan, the number of expanded states, and the time. While the search and the animation run, the move buttons are disabled, so the user cannot change the cube under the solver. Figure 3 shows the full window with a scrambled cube.



Figure 3: The simulator window. The cube net is on the left, the solver log on the right, the twelve move buttons and the solve button below, with a menu to pick the strategy.

4 The A* algorithm

4.1 Principle

A* is an informed search. It expands the node with the smallest evaluation function

$$f(n) = g(n) + h(n),$$

where $g(n)$ is the cost from the start to n , and $h(n)$ estimates the cost from n to the goal. When h is admissible, that is, when it never overestimates the real cost, A* returns an optimal solution. With $h = 0$ the function f becomes $f = g$, so A* behaves as a uniform cost search.

The course presents several search strategies. Table 4 compares the ones we use. The uniform cost search is complete and optimal but blind. It explores by cost only. A* keeps the same guarantees but adds the heuristic, so it looks toward the goal. This is why A* expands far fewer states when the heuristic is good.

Strategy	Orders the frontier by	Optimal	Uses a heuristic
Breadth first search	depth	yes (unit cost)	no
Uniform cost search	g	yes	no
Greedy best first	h	no	yes
A*	$f = g + h$	yes (admissible h)	yes

Table 4: Search strategies of the course. We use uniform cost search and A*.

Figure 4 shows a small search tree. Each node carries its values g , h and f . A* always expands the leaf with the smallest f . A node that looks cheap by g alone may be skipped if its heuristic is large, and the other way around.

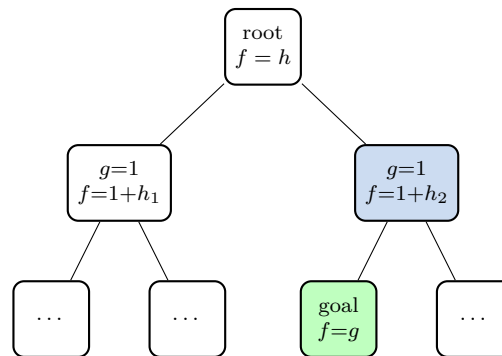


Figure 4: A small A* search tree. The frontier is ordered by $f = g + h$. The shaded node has the smallest f , so it is expanded first and reaches the goal.

4.2 The solve method

The `solve` method follows the A* pseudo code of Lecture 2. It uses the frontier and the explored set. It tests the goal when a node is popped, which keeps the search optimal. It rebuilds the plan by walking the parent chain and reading the action of each node. Listing 2 shows the algorithm.

Listing 2: The A* search loop, as implemented in `solve`.

```

push the root state into the frontier
while the frontier is not empty:
    node = frontier.pop()           # smallest f = g + h
    if node already expanded with a smaller g: continue
    add node to the explored set
    if node is the goal: return the plan (walk the parent chain)
    for each child in node.expand(): # the 12 turns
        if child already expanded with a smaller g: continue
        frontier.push(child)
return failure
  
```

4.3 Reconstructing the solution

When the goal is found, we follow the parent links from the goal to the root. At each step we read the action that produced the node. We reverse the order. This gives the list of moves from

the start to the solved cube. The `solve` method returns this list as strings, such as "R" or "U".

4.4 Why A* is optimal

The course states that A* is optimal when the heuristic is admissible. The idea is simple. A* tests the goal only when a node leaves the frontier with the smallest f . Suppose this goal node has cost g . Because the goal is reached, h is zero there, so $f = g$. Any other node n still in the frontier has $f(n) = g(n) + h(n)$. Since h never overestimates, $f(n)$ is a lower bound on the cost of any solution through n . As n was not chosen, $f(n) \geq g$. So no cheaper solution can exist. The first goal popped is therefore optimal. We also keep an explored set, so a configuration reached again by a longer path is skipped, which is the graph version of A* from the course.

4.5 Data structures

The frontier is a priority queue ordered by f . Java does not offer a cheap way to lower the key of an element already in the queue. So when a better path to a configuration appears, we push the new entry and mark the old one as out of date. Out of date entries are skipped when they are popped. The explored set is a hash map from a configuration to its best known cost. Both structures compare states by their cube, so they detect repeated configurations and keep the search from looping.

5 Heuristics

A heuristic must be admissible to keep A* optimal. Both of our heuristics come from a relaxation of the problem, as in the tutorial on informed search. A relaxation removes some rules, so its optimal cost is a lower bound on the real cost.

5.1 HeuristicCube: the piece relaxation

This heuristic assumes that each piece moves on its own. A piece is a corner with three stickers or an edge with two stickers. The six centers never move. A piece is well placed when all of its stickers are on their target faces. A single turn moves four corners and four edges at once. So one turn can fix at most four corners and at most four edges. The number of remaining moves is therefore at least

$$h_{\text{cube}} = \max \left(\left\lceil \frac{\text{misplaced corners}}{4} \right\rceil, \left\lceil \frac{\text{misplaced edges}}{4} \right\rceil \right).$$

The maximum of two lower bounds is still a lower bound, so this heuristic is admissible.

5.2 HeuristicLabel: the sticker relaxation

This heuristic assumes that each sticker moves on its own. A sticker only needs to reach its correct face. We measure the 3D Manhattan distance from its face to its target face, counted in quarter turns. The distance is 0 on the correct face, 1 on an adjacent face, and 2 on the opposite face. We sum this distance over all 54 stickers. We then divide by the number of stickers that can change face in one turn, which is twelve.

$$h_{\text{label}} = \frac{1}{12} \sum_{s=1}^{54} d(s).$$

One turn moves twelve belt stickers. Each one moves to an adjacent face, so it cuts its distance by at most one. The total distance drops by at most twelve per move. The number of remaining moves is therefore at least the sum divided by twelve. So this heuristic is admissible.

5.3 Checking admissibility

We checked both heuristics by tests. On a cube scrambled with k random moves, the heuristic must not exceed k , because undoing the scramble takes at most k moves. We ran this check for many seeds and depths. The heuristic value was always at most k . We also checked that both heuristics return zero on the solved cube. A stronger check appears in the experiments: A* with these heuristics returns the same solution length as the uniform cost search, which proves that they keep the search optimal.

5.4 A small worked example

Take a solved cube and apply one move, the turn F. This move sends twelve belt stickers to an adjacent face. For the sticker heuristic each of these twelve stickers now sits on a face next to its target, so its distance is one. The sum is twelve, and $h_{\text{label}} = 12/12 = 1$. The true cost is also one, so the estimate is exact here. For the piece heuristic this same move misplaces four corners and four edges. So the value is $\max(\lceil 4/4 \rceil, \lceil 4/4 \rceil) = 1$. Both heuristics agree with the real cost of one move.

5.5 Comparing the two heuristics

A heuristic h_a dominates another h_b when $h_a(n) \geq h_b(n)$ for every state n , while both stay admissible. A dominating heuristic is never worse, and it usually expands fewer states. In our tests the sticker heuristic gives larger values on most scrambled cubes, so it guides the search better. The experiments in Section 6 confirm this: the sticker heuristic expands fewer states than the piece heuristic at the same depth. Both remain admissible, so both keep A* optimal.

6 Experiments

6.1 Setup

We measured three strategies. The first is the uniform cost search, where $h = 0$. The second is A* with the piece heuristic. The third is A* with the sticker heuristic. For each strategy and each scramble depth, we solved several cubes. We recorded the success rate, the average number of expanded states, the average time, and the average solution length. A run fails when it expands more states than a fixed limit. The seeds are fixed, so the numbers can be reproduced.

6.2 Results

Table 5 reports the main numbers. The pattern is clear. The uniform cost search blows up around depth seven. It expands hundreds of thousands of states at depth six. It then fails at depth seven and eight under the limit. Both heuristics scale much further. The sticker heuristic is the strongest here. It expands far fewer states than the piece heuristic.

Depth	UCS ($h = 0$)		A* HeuristicCube		A* HeuristicLabel	
	expanded	time (ms)	expanded	time (ms)	expanded	time (ms)
4	6 818	26	8	0.1	14	0.2
5	20 599	86	64	0.7	40	0.9
6	503 575	3 170	450	5.2	400	5.7
7	fails	n/a	6 953	66	1 597	16
8	fails	n/a	26 304	207	5 793	56

Table 5: Average expanded states and time per strategy and scramble depth. UCS fails from depth seven under the node limit.

Figure 5 draws the number of expanded states on a logarithmic scale. The gap between the uniform cost search and A* grows very fast. At depth six the difference is about three orders of magnitude. The two A* curves stay low and close, with the sticker heuristic slightly below the piece heuristic.

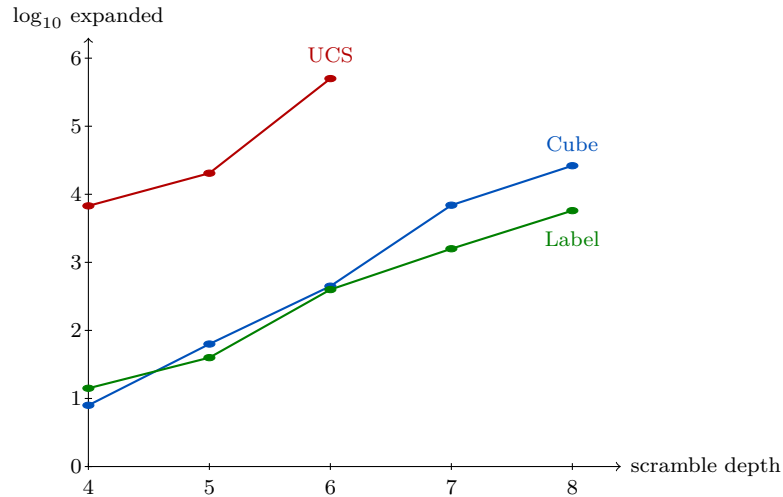


Figure 5: Expanded states by depth, log scale. UCS grows much faster and stops at depth six within memory. Both A* curves stay low.

6.3 A complete example

To make the result concrete, we scramble a solved cube with seven moves and solve it with the sticker heuristic. Table 6 shows the scramble, the plan, and the search effort. The plan has seven moves, so it is optimal. We can also see that the plan is the scramble read backward, with each move replaced by its inverse. This is the shortest way to undo the scramble, and A* finds it.

Scramble (7 moves)	F U' L U R' D L'
Solution (7 moves)	L D' R U' L' U F'
Expanded states	1 110
Generated states	12 030
Time	about 20 ms

Table 6: A complete solve with the sticker heuristic. The plan undoes the scramble in the optimal number of moves.

6.4 Optimality

The solution length is the same for the three strategies at every depth. For example, the average length is 6.0 at depth six and 7.6 at depth eight, for all three. This is an empirical proof that the heuristics are admissible. A heuristic that overestimated the cost would make A* return a longer plan than the uniform cost search. This never happens.

6.5 Discussion

The experiments confirm the course. First, the heuristic does not change the answer, only the effort to find it. The plans stay optimal. Second, a good heuristic gives huge savings. At depth six the uniform cost search expands about 500 000 states, while A* expands a few hundred. Third, the sticker heuristic dominates the piece heuristic here, because it gives finer values. Finally, A* runs out of memory before it runs out of time, exactly as the slides state. The limit we hit first is the number of stored states, not the clock.

6.6 A simple optimization

We also added one optional optimization. After a move, we never apply its inverse right away. For example we never play U' just after U. Such a pair cancels out, so it can never belong to an optimal solution. Skipping it keeps the search optimal and makes the tree smaller. This option is turned off by default, so the `solve` method stays close to the pseudo code of the course. When it is turned on, A* expands fewer states at the same depth.

6.7 Methodology

We built the project with a test first method. For each class we wrote the tests before the code. We made the tests fail, then wrote the code until they passed. The cube tests check the algebraic invariants. The search tests check that the plan really solves the cube and that A* matches the optimal length. In total the project has twenty four tests, and they all pass. The code was also reviewed, which helped us find and fix a rotation that turned the wrong way and a few performance issues.

6.8 Limits

A* is still exponential in the worst case. Our solver stays fast up to about eight or nine random moves. Beyond that, the frontier grows too large for memory. This is a known limit of A* on the Rubik's Cube. Stronger methods exist, but they are outside the scope of this lab.

7 Conclusion

We built a program that solves the Rubik's Cube with the A* algorithm. We modeled the cube as a search problem with the five components of the course. We followed the statement and built the classes `Cube`, `State`, `Astar`, `StateSet`, `PriorityQueueState`, and the heuristics. We first ran a uniform cost search. We then added two admissible heuristics, both from a relaxation of the problem.

The results match the theory. A* returns an optimal solution. A good heuristic cuts the number of expanded states by several orders of magnitude. The uniform cost search fails from about seven

random moves, while A* solves more scrambled cubes. The main limit is memory, not time. This project gave us a clear, hands on view of informed search and of why heuristics matter.

Team contributions

The three members worked together on the design and the report. The work was shared as follows. One member focused on the cube model and the turns. One member focused on the A* search and the data structures. One member focused on the two heuristics and the experiments. The interface, the tests, and the writing were a shared effort, with regular reviews between the members.